

Cotard's syndrome: analysis of 100 cases

Berrios GE, Luque R. Cotard's syndrome: analysis of 100 cases.
Acta Psychiatr Scand 1995; 91: 185–188. © Munksgaard 1995.

In 1880, Jules Cotard reported a clinical state he believed was a new type of agitated melancholia. A statistical analysis has been carried out of 100 cases of Cotard's syndrome to determine how this clinical concept has fared since its inception. In terms of clinical profile, no difference was found between men and women or between underlying diagnostic categories; age seemed to increase the likelihood of developing *délire des négations*. Depression was reported in 89% of subjects; the most common nihilistic delusions concerned the body (86%) and existence (69%). Anxiety (65%) and guilt (63%) were also common, followed by hypochondriacal delusions (58%) and delusions of immortality (55). An exploratory factor analysis extracted 3 factors: psychotic depression, Cotard type I and Cotard type II. The psychotic depression factor included patients with melancholia and few nihilistic delusions. Cotard type I patients, on the other hand, showed no loadings for depression or other disease and are likely to constitute a pure Cotard syndrome whose nosology may be closer to the delusional than the affective disorders. Type II patients showed anxiety, depression and auditory hallucinations and constitute a mixed group. This new grouping cuts across the more traditional view and may have therapeutic implications. Authors, in general, have considered *délire des négations* as a syndrome rather than a new disease and do not seem to support the view that the completeness of the syndrome is a function of presence or severity of depression. The view that *délire des négations* refers only to the delusion of being dead has also carried little favour as it is likely to waste information.

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Key words: depression; delusion; Cotard

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Accepted for publication October 29, 1994

There has been some recent interest in English-speaking psychiatry in the clinical (1–3) and neurobiological (4, 5, 6) aspects of Cotard's syndrome, and the tendency to use, irrespective of clinical context, Cotard's delusion for the delusional belief of being dead (4, 6). There is little evidence that such departure from the original description is justified by the historical record. A detailed analysis of the original sources and of the conceptual construction of the *Délire de Cotard* (Cotard's syndrome) has appeared elsewhere (7).

In 1880, Jules Cotard (8) reported the case of a 43-year-old woman who believed that she had “no brain, nerves, chest, or entrails, and was just skin and bone”, that “neither God or the devil existed”, and that she did not need food for “she was eternal and would live forever”. Cotard diagnosed her as suffering from *lypémanie*, an Esquirolean category only partially related to psychotic depression episode (9); and explained that this *délire hypochondriacque* resulted from “an interpretation of pathological sensations often present in patients with anxious melancholia”. Cotard believed that he had identified a new type of depression characterized by anxious

melancholia, ideas of damnation or possession, suicidal behaviour, insensitivity to pain, delusions of nonexistence and of immortality. Two years later, Cotard (10) referred to the same clinical state as *délire des négations* (nihilistic syndrome). This was wrongly rendered into English as nihilistic delusion (for a discussion on the untranslatability of the French concept of *délire* see Berrios & Luque (7)).

Soon after Cotard's death the debate started as to whether he had actually described a new disease or just a severe form of melancholia (7). Régis (11) added a third possibility that Cotard had described a syndrome that could accompany mental disorders other than melancholia and coined the eponym *délire de Cotard*. This view proved influential, and the French eponym was rendered into English as Cotard's syndrome.

Tissot (12) proposed that the *délire des négations* included an affective component (anxiety) and an intellectual component (the ideas of negation); and Loudet & Martinez (13) identified two types: one was an isolated, nonsystematized *délire des négations* associated with general paralysis, alcoholic psychoses and dementia; and the second was a system-

atized state (or true Cotard's syndrome) only found in anxious melancholia and chronic hypochondria.

Interest in the Cotard state was renewed after the Second World War. Perris (1955) added a fourth interpretation by contending that Cotard's intention had actually been to describe a single symptom, a hypochondriacal delusion occurring in anxious melancholia (14). Others resuscitated the earlier view that Cotard's syndrome might, after all, be a different entity (5, 15, 16).

In summary, authors have mostly considered the Cotard's state to be either a new disease or a syndrome. To ascertain which has been the predominant view it would seem necessary to determine the patterns of clinical usage since Cotard first described his *délire*. This article statistically analyzes 100 cases of Cotard's syndrome reported since 1880.

Material and method

To obtain the sample, more than 200 papers containing reports of cases of Cotard's syndrome were perused. A data sheet was used to collect the following 19 variables: age, sex, presence of anxiety, depression, nihilistic delusions (concerning: body, existence or concepts), delusions (of: hypochondriacal concern, immortality, guilt or damnation), other delusions, visual and auditory hallucinations, negativism, suicidal ideas and presence of organic brain disorder (as diagnosed by the original authors). Two raters (independently) diagnosed each case as: major depression, bipolar affective disorder, schizophrenia, organic disorder or other. Lastly, a Cotard index was constructed by totalling the number of nihilistic delusions and for practical purposes was considered as a measure of syndrome completeness. The index was then used to divide the case (for exploratory purposes) into complete (score = 3) and incomplete (score < 3).

By this procedure, 100 complete cases were collated from the literature (17–59). There was no reason to believe that the incomplete or partial reports *perforce* left out from the series included cases that where different or atypical in any way. Since the exclusive aim of this study was to construct a phenomenological map of historical clinical usage and not to make absolute claims as to the general clinical attributes of the syndrome, it was felt that there was no need to use control samples. Statistical analysis was carried out by means of the Statistical Package for the Social Sciences, Version 2.0 (60).

Results

The mean age for the sample was 52 years ($SD = 14.5$, range = 16 to 81, mode and median = 53). The age distribution is shown in Fig. 1, and frequency of

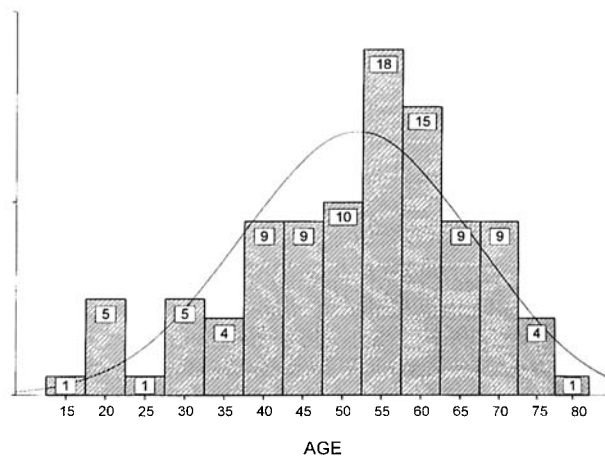


Fig. 1. 100 cases of Cotard's syndrome. Distribution by age: numbers on bars are absolute.

symptoms in Fig. 2. No difference was found between male and female patients except on the presence of organic disorder which was more frequent in females (Mann-Whitney (M-W), $P < 0.001$). Age was found to increase the probability of having hypochondriacal delusions ($P < 0.02$), depression ($P < 0.005$), nihilistic delusions concerning the body ($P < 0.004$) and the nonexistence of the self ($P < 0.03$).

Two subgroups were then created according to whether the Cotard index was complete (38 cases) or incomplete (62 cases); comparison showed that the complete group showed more often anxiety (M-W, $P < 0.02$) and visual hallucinations (M-W, $P < 0.04$).

Patients were then compared according to clinical diagnosis by means of a nonparametric one-way analysis of variance (Kruskal-Wallis). The depressive sample was found to show more anxiety ($P < 0.01$), guilt ($P < 0.02$), depression ($P < 0.001$) and suicidal behaviour ($P < 0.01$); the organic sample

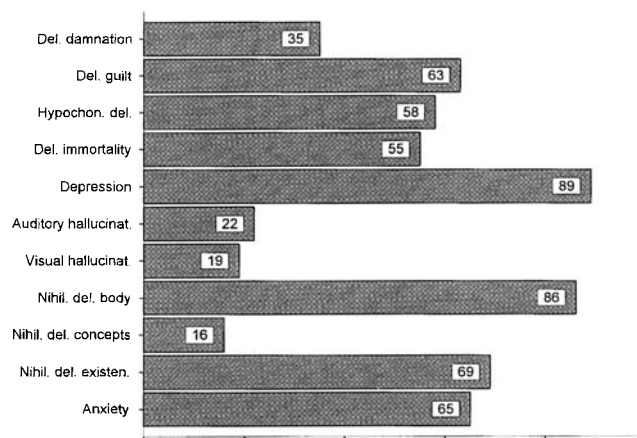


Fig. 2. 100 cases of Cotard's syndrome. Frequency of symptoms: numbers on bars are absolute.

was found to be the oldest ($P < 0.01$) and the schizophrenic sample to have more auditory hallucinations ($P < 0.001$). There was no difference on the Cotard index between the diagnostic groups.

Lastly, data were summarized by means of an exploratory factor analysis. This yielded 3 factors: factor 1: psychotic depression (eigenvalue: 3.13; 18% variance) with loadings from age (0.40), anxiety (0.47), delusions of guilt (0.52), depression (0.61), and auditory hallucinations (0.66); factor 2: Cotard type I (eigenvalue 2.16; 13% variance) with loadings from age (0.54), hypochondriacal delusions (0.66), nihilistic delusions concerning body (0.75), concepts (0.47) and existence (0.59); and factor 3: Cotard type II (eigenvalue 1.83, 11% variance) with loadings from anxiety (0.35), delusions of immortality (0.86), auditory hallucinations (0.35), nihilistic delusions concerning existence (0.37) and suicidal behaviour (0.59).

Discussion

This study first offered a short account of the conceptual history of what has come to be known as Cotard's syndrome and a statistical analysis of 100 cases taken from the rich historical database that has accumulated since 1880. It has not been its aim to inquire into the incidence, prevalence, aetiology and absolute clinical attributes of the Cotard syndrome but to map historical usage in western psychiatry practice.

In spite of the varied provenance of the sample, the analysis has yielded important information on the way in which Cotard's intentions were interpreted after his death. The clinical state he described has been reported in all age groups, including a subject as young as 16. In this regard, authors do not seem to have paid heed to his belief that the condition was a form of agitated melancholia affecting people in middle age or later.

Analysis of the sample in terms of the Cotard index also showed that no important difference was found between men and women nor between the four diagnostic categories; this latter finding suggests that there has been a tendency to consider Cotard's state as a syndrome not particularly connected with one type of psychosis. However, analysis of the frequency of symptoms showed that, in spite of this syndromatic view, depression was nevertheless reported in 89% of subjects. Among the nihilistic delusions, the commonest were those concerning the body (86%) and existence (69%). Anxiety (65%) and guilt (63%) were also common followed by hypochondriacal delusions (58%) and delusions of immortality (55%).

The exploratory factor analysis was able to extract a psychotic depression factor and two Cotard fac-

tors that, for lack of a better term, were labelled types I and II. The psychotic depression factor included patients with melancholia and only few nihilistic delusions. Cotard type I patients, on the other hand, showed no loadings for depression or other disease and included most of the complete cases (on the Cotard index); these patients are likely to constitute a pure Cotard syndrome, and it is suggested here that their nosological origin is in the delusional and not in the affective disorders; the therapeutic implication of this view is that such patients are less responsive to antidepressant treatment. Type II patients showed anxiety, depression and auditory hallucinations and constitute a mixed group.

In general, the authors reviewed here tend to consider that the completeness of the Cotard syndrome is not a specific function of the presence or severity of depression. Likewise, there is little support for the view that *délire des négations* should refer only to the delusion of being dead. Such a view is likely to waste information and hamper any possibility of finding out whether the symptom-cluster involved has any stable brain representation.

The next step is to carry out a similar analysis in a prospective sample so that covariation between the completeness of the Cotard syndrome and the severity of underlying disease can be assessed and the natural history of its components mapped. Only after this is done will it be possible to determine whether it is worth bothering about the clinical and prognostic implications of the Cotard syndrome or its neurobiology.

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